COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE212 | Automating Server Management	**SECTION:**	CPE31S1
DATE PERFORMED:	9-24-26	**DATE SUBMITTED:**	9-24-26
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ACTIVITY NO. – **Midterm Skills Exam**

1. PROCEDURE OUTPUT

Task 1: Prepare the Server
Dependencies:

```
  GNU nano 7.2              roles/prometheus/tasks/ubuntu.yml
---
- name: Install Prometheus dependencies on Ubuntu
  ansible.builtin.apt:
    name:
      - wget
      - tar
    state: present
```

- Installs dependencies

Task 2: Create the Prometheus User

```
- name: Create Prometheus system user
  ansible.builtin.user:
    name: prometheus
    system: yes
    shell: /bin/false
    create_home: no
```

- Creates users

Task 3: Create Prometheus Directories and config files

```
- name: Create Prometheus config directory
  ansible.builtin.file:
    path: /etc/prometheus
    state: directory
    owner: prometheus
    group: prometheus
    mode: '0755'
```

```
- name: Create Prometheus data directory
  ansible.builtin.file:
    path: var/lib/prometheus
    state: directory
    owner: prometheus
    group: prometheus
    mode: '0755'
```

Creates directories

Task 4: Download Prometheus

```
- name: Download Prometheus
  ansible.builtin.get_url:
    url: https://github.com/prometheus/prometheus/releases/download/v3.14.0/pro>
    dest: /tmp/prometheus-3.14.0.linux-amd64.tar.gz
    mode: '0644'
```

- Downloads prometheus

Task 5: Install the Prometheus Binaries

```
- name: Install Prometheus' binary
  ansible.builtin.copy:
    src: /tmp/prometheus-3.14.0.linux-amd64/prometheus
    dest: /usr/local/bin/prometheus
    remote_src: yes
    owner: root
    group: root
    mode: '0755'
```

- Downloads binaries

Task 6: Install the Prometheus Configuration Files

```
- name: Create Prometheus config directory
  ansible.builtin.file:
    path: /etc/prometheus
    state: directory
    owner: prometheus
    group: prometheus
    mode: '0755'
```

- Creates config files

Task 7: Validate the Configuration

```
- name: Validating Prometheus Config
  ansible.builtin.command:
    cmd: /usr/local/bin/promtool check config /etc/prometheus/prometheus.yml
  changed_when: false
```

- Validates config files

Task 8: Create the systemd Service

```
- name: Create Prometheus Systemd Service
  ansible.builtin.copy:
    dest: /etc/systemd/system/prometheus.service
    owner: root
    group: root
    mode: '0644'
    content: |
      [Unit]
      Description=Prometheus Monitoring
      Wants=network-online.target
      After=network-online.target

      [Service]
      User=prometheus
      Group=prometheus
      Type=simple
      ExecStart=/usr/local/bin/prometheus --config.file=/etc/prometheus/prometh>
      Restart=on-failure

      [Install]
      WantedBy=multi-user.target
```

Creates systemd

Task 9: Start and Enable Prometheus

```
- name: Enable and start Prometheus
  ansible.builtin.systemd:
    name: prometheus
    enabled: yes
    state: started
```

Starts and enables sytemd

Task 10: Verify Prometheus

```
× prometheus.service - Prometheus Monitoring
     Loaded: loaded (/etc/systemd/system/prometheus.service; enabled; preset: enabled)
     Active: failed (Result: exit-code) since Thu 2026-09-24 02:18:21 UTC; 17s ago
   Duration: 160ms
    Process: 25497 ExecStart=/usr/local/bin/prometheus --config.file=/etc/prometheus/prometheus.ym>
   Main PID: 25497 (code=exited, status=1/FAILURE)
        CPU: 198ms

Sep 24 02:18:21 server2 systemd[1]: prometheus.service: Scheduled restart job, restart counter is >
Sep 24 02:18:21 server2 systemd[1]: prometheus.service: Start request repeated too quickly.
Sep 24 02:18:21 server2 systemd[1]: prometheus.service: Failed with result 'exit-code'.
Sep 24 02:18:21 server2 systemd[1]: Failed to start prometheus.service - Prometheus Monitoring.
~
~
```

Verifies prometheus

Task 11: Access Prometheus Through a Browser



2. ANSWERS TO SUPPLEMENTAL ACTIVITY

CONCLUSION:
I have learned a lot from this Midterm exam, mainly how to create an ansible playbook to install prometheus. With the use of ansible.builtin commands, I was able to create a playbook that is able to install requirements, config files, directories, and much more for Prometheus. I was only able to finish ubuntu.yml but then encountered a failure in the end and only start on centos.yml But all in all, I was able to understand how ansible works and I hope to use this in future HOAs.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

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Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

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